BHAVESH SABNANI

Chandigarh, India | 9664320613 | bhavesh.sabnani2005@gmail.com
linkedin.com/in/bhavesh-sabnani-256946372 | github.com/BhaveshSab | bhaveshsabnani.tech

SUMMARY

Software engineering student (B.Tech, PEC, graduating May 2028) with strong applied experience building, breaking and rebuilding full-stack web applications in React, Node.js, Python, C++ and SQL, front-end and back-end. Builds high-volume, scalable, payment-style workflows (approvals, purchase orders, invoices) with exact money handling, REST APIs, concurrency control and 525 test assertions; has estimated, designed, tested and triaged real issues (race conditions, a bundle-loading bug, a rate-limiter bug), and works with cross-functional teams as a collaborator and team leader.

EDUCATION

Punjab Engineering College (PEC), Chandigarh | B.Tech, Electrical Engineering | GPA: 7.69/10 | Aug 2024 - May 2028 (Expected)
JEE Mains: 98.04 percentile (1.6M+ candidates)
XII, RBSE Board, Jaipur | 87% (90% PCM & English) | 2023
X, RBSE Board, Jaipur | 92% | 2021

WORK EXPERIENCE

Full Stack Developer Intern - Kagnus Engineering Network | Remote Internship Oct 2026 - Nov 2026
VendorBridge ERP | Repo: github.com/BhaveshSab/VendorBridge-kagnus | Live: vendor-bridge-kagnus.vercel.app

- **Full-stack:** Built a React 19/TypeScript single-page app (HTML/DOM, CSS, 22 lazy-loaded pages, TanStack Query caching, AJAX calls through one typed Fetch wrapper) on a Node.js/Express 5 REST API with 51 endpoints and a 19-model PostgreSQL schema (Prisma); 4 role-based tiers.
- **Payments-style workflow:** Built the chain from RFQ to quotation comparison, approvals, purchase orders and invoices; money computed server-side as fixed-point DECIMAL(14,2) with half-up rounding; multi-currency spend bucketed per currency, never summed across them.
- **Concurrency & reliability:** Made approval-to-invoice race-safe with `SELECT ... FOR UPDATE` row locks and unique indexes (one decision, one document); idempotent purchase-order generation; 8 database triggers and 9 CHECK constraints; errors mapped to stable status codes (422, 503 with Retry-After).
- **Testing & security:** 5 Node test harnesses with 525 assertions (227 API checks on record); httpOnly-cookie JWT sessions, vendor-scoped queries, 25 Zod request schemas, fail-open Redis rate limiting that survives restarts and multi-instance deploys.

PROJECTS

AssetFlow - Asset & Resource Management ERP | NestJS, PostgreSQL, Redis, React 18, TypeScript | Repo: github.com/BhaveshSab/assetflow | Live: odoohackathon-ujkn.vercel.app

- **Back-end:** NestJS REST API with 78 endpoints, a 13-model PostgreSQL schema and JWT RBAC across 4 roles; out-of-scope records return 404, not 403, so IDs cannot be probed; allocations and bookings take a row lock before their conflict check; invalid transitions return 409.
- **Distributed rate limiting:** Redis counters with an atomic Lua increment make the limit global across instances, and fail open to in-process counters if Redis is down, surfaced as a Prometheus gauge and alert instead of an outage.
- **Front-end:** React 18/TypeScript app with 12 lazily-loaded views (9 screens + 3 role dashboards); one typed API client for 76 call sites with a 20 s timeout and a single 401 path; first-load JavaScript cut from 294 kB to 144 kB gzipped after tracing a charts-chunk bug.
- **Testing:** 256 automated checks (72 unit, 117 e2e on real PostgreSQL via PGlite, 67 frontend); contrast script fails the build below WCAG AA; ESLint gate with zero `any`.

AssetFlow - Self-Hosted Infrastructure | AWS EC2, Docker Compose, Nginx, Prometheus, Grafana, GitHub Actions | Infra: 13.55.90.25

- Deployed on AWS EC2 with Docker Compose (10 services) behind Nginx with Let's Encrypt TLS; nightly pg_dump to encrypted S3, restore verified in 14 s; CI runs lint, shellcheck and 256 tests; failed deploys roll back automatically.
- Triaged a production bug: the rate limiter was keyed on client IP, so everyone behind one address shared 300 req/min; re-keyed it to the user. Monitoring: Prometheus (11 alert rules) to Alertmanager to Telegram, JSON logs with request-ID tracing.

AI-Driven Smart Grid Node & Cloud Telemetry Pipeline | C++, Python, ESP32, MQTT, Scikit-Learn, Streamlit

- ESP32 node streams 230V AC telemetry over MQTT; a C++ state machine enforces load limits and a Python server applies an Isolation Forest model to flag abnormal appliance stacking; Streamlit dashboard shows live data.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, SQL
Web Technologies: HTTP, HTML5/DOM, CSS3, AJAX (Fetch API), JSON, REST APIs, web services (REST), Responsive Design
Front-end: React, TypeScript, TanStack Query, Tailwind CSS, Accessibility (WCAG 2.1 AA)
Back-end & Databases: Node.js, Express, NestJS, MySQL, PostgreSQL, Redis, Prisma, Zod, JWT/RBAC
Distributed Systems & Scale: Concurrency control (row locking, transactions, idempotency), Redis caching and distributed rate limiting, lazy loading and code splitting
Testing & DevOps: Unit, functional, integration, API and end-to-end testing, debugging, Postman, Git/GitHub, GitHub Actions CI/CD, Docker, AWS (EC2, S3), Nginx, Linux, Prometheus, Grafana
Computer Science Fundamentals: DSA (250+ LeetCode), OOP (C++), DBMS, Operating Systems, Computer Networks, SDLC; Agentic AI (RAG, MCP)

CERTIFICATIONS & ACHIEVEMENTS

- Redis Associate Developer (Redis, Aug 2026, score 92%)
- Oracle Agentic AI Certified Foundations Associate (Oracle, Jul 2026)
- Programming in Modern C++, Elite (NPTEL / IIT Kharagpur, 2025, top 5% of 26,183)
- Kagnus Internship Program, Certificate of Completion (Nov 2026) | ID: KGN-CERT-26-FS-G38541 | talent.kagnus.com/verify
- Smart India Hackathon (SIH): Team leader; collaborated with teammates to get selected for the college internal round, ranking in the top tier out of 100+ teams.